# TWO CATEGORICAL REFORMULATIONS OF THE CONDITION (RS)

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-defos.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the derivation of the 2-fibre product description of the comparison functor from (RS), the trivial implication, and the explicit computation of morphism sets recovering the fibre product description in the small extension case.

*Editorial note.* The label lemma-RS-2-categorical names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source formal-defos.tex –label lemma-RS-2-categorical –start 5093 –end 5150 –local-id 28018 returned RESULT=OK (L28018 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-formal-defos/candidates-halfA.json.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Description of objects of the fibre product category. Same chapter, the lemma that every object of the category over a fibre product arises as a fibre product of objects over the two factors. Supplies the description of the objects of F(A_1 x_A A_2) as x_1 x_x x_2 and of the morphism sets.
- Categorical description of the 2-fibre product. Categories, the remark describing the 2-fibre product of categories through objects and morphisms. Supplies the conclusion that the comparison functor is an equivalence, i.e. that F(A_1 x_A A_2) is a 2-fibre product.
- Reduction of (RS) to small extensions. Same chapter, the lemma that (RS) for all surjections follows from (RS) for small extensions. Supplies the final step passing from the small extension case to all surjections.

*Source reference.* Original remark 06RL: sources/stacks-chow/tags.txt, source lines 8645–8645. Prerequisite pointer invoked at proof line(s) 5120; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original equation 05NN: sources/stacks-formal-defos/pinned-original-source.tex, source lines 2822–2822. Prerequisite pointer invoked at proof

*Date*: stacks-project:formal-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

line(s) 5102; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 06J3: sources/stacks-formal-defos/pinned-original-source.tex, source lines 5036–5036. Prerequisite pointer invoked at proof line(s) 5110; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original reference 06J4: sources/stacks-formal-defos/pinned-original-source.tex, source lines 5081–5081. Prerequisite pointer invoked at proof line(s) 5149; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $\mathcal{F}$ be a category cofibered in groupoids over $\mathcal{C}_\Lambda$. The following are equivalent*

1. *$\mathcal{F}$ satisfies (RS),*
2. *the functor $\mathcal{F}(A_1 \times_A A_2) \to \mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$ see (05NN) is an equivalence of categories whenever $A_2 \to A$ is surjective, and*
3. *same as in (2) whenever $A_2 \to A$ is a small extension.*

*Proof.* Assume (1). By Lemma 06J3 we see that every object of $\mathcal{F}(A_1 \times_A A_2)$ is of the form $x_1 \times_x x_2$. Moreover

$$\mathrm{Mor}_{A_1 \times_A A_2}(x_1 \times_x x_2, y_1 \times_y y_2) = \mathrm{Mor}_{A_1}(x_1, y_1) \times_{\mathrm{Mor}_A(x,y)} \mathrm{Mor}_{A_2}(x_2, y_2).$$

Hence we see that $\mathcal{F}(A_1 \times_A A_2)$ is a 2-fibre product of $\mathcal{F}(A_1)$ with $\mathcal{F}(A_2)$ over $\mathcal{F}(A)$ by Categories, Remark 06RL. In other words, we see that (2) holds.

The implication (2) $\Rightarrow$ (3) is immediate.

Assume (3). Let $q_1 : A_1 \to A$ and $q_2 : A_2 \to A$ be given with $q_2$ a small extension. We will use the description of the 2-fibre product $\mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$ from Categories, Remark 06RL. Hence let $y \in \mathcal{F}(A_1 \times_A A_2)$ correspond to $(x_1, x_2, x, a_1 : x_1 \to x, a_2 : x_2 \to x)$. Let $z$ be an object of $\mathcal{F}$ lying over $C$. Then

$$\begin{aligned}
\mathrm{Mor}_{\mathcal{F}}(z, y) &= \{(f, \alpha) \mid f : C \to A_1 \times_A A_2, \alpha : f_* z \to y\} \\
&= \{(f_1, f_2, \alpha_1, \alpha_2) \mid f_i : C \to A_i,\ \alpha_i : f_{i,*} z \to x_i, \\
&\qquad q_1 \circ f_1 = q_2 \circ f_2,\ q_{1,*}\alpha_1 = q_{2,*}\alpha_2\} \\
&= \mathrm{Mor}_{\mathcal{F}}(z, x_1) \times_{\mathrm{Mor}_{\mathcal{F}}(z,x)} \mathrm{Mor}_{\mathcal{F}}(z, x_2)
\end{aligned}$$

whence $y$ is a fibre product of $x_1$ and $x_2$ over $x$. Thus we see that $\mathcal{F}$ satisfies (RS) in case $A_2 \to A$ is a small extension. Hence (RS) holds by Lemma 06J4. $\square$